

LipCoordFormer: A Transformer-Based Lip-Reading Model with Visual-Landmark-Temporal Fusion

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Abstract

Lip-reading translation refers to converting visual lip movements into human-understandable natural sentences. Current state-of-the-art lip-reading approaches are based on frame-level spatiotemporal convolutions and recurrent networks, which struggle with visual noise from non-lip regions, inadequate modeling of long-range temporal dependencies, and lacks visual-geometric alignments. To address these limitations, we propose LipCoordFormer, a transformer-based architecture integrating spatiotemporal image encoder and lip movement landmarks to enable joint learning of visual-geometric patterns while eliminating frame-level alignment dependencies. Our model achieves a character error rate (CER) of 2.78% and word error rate (WER) of 7.39% on the GRID corpus, which is comparable to the state-of-the-art models trained on the same dataset. (LipNet: CER: 2.0%, WER: 4.8%; LCArNet: CER: 1.3%, WER: 2.9%; STFM: WER: 1.3%) while introducing critical advancements in noise robustness and temporal dependency modeling. This result is achieved using only 1/4 training data of the current approaches and further reduce training time to 12 hours.

Keywords— Lip Reading, Recurrent Networks, Spatiotemporal Convolutions, Transformer Decoder, Lip Movement Landmarks

1. Introduction

The ability to translate lip movements into text represents a transformative advancement in the field of Artificial Intelligence, particularly in the realm of communication accessibility. This research focuses on enhancing LipNet [1], an innovative system designed to perform end-to-end sentence-level lip reading. LipNet employs Spatiotemporal Convolutions (STCNNs), Gated Recurrent Units (GRUs), and Connectionist Temporal Classification (CTC) loss to achieve robust performance in recognizing spoken language through visual cues alone.

Despite LipNet being a popular model in lip-reading task, its accuracy remains constrained by three fundamental limitations:

(1) subjective to visual noise from non-lip facial regions [2], (2) inadequate modeling of long-range temporal dependencies [3], and (3) reliance on frame-level alignment heuristics rather than learned linguistic features [4]. The societal motivation behind improving LipNet stems from its potential to significantly aid individuals with hearing impairments, enabling them to engage more fully in conversations and understand spoken language in various environments. Furthermore, this technology has promising applications in security systems, where silent speech recognition can facilitate communication in noisy settings or during situations that require discretion.

To advance LipNet’s capabilities, we propose exploring several strategies.

- **Image Preprocessing:** By removing distracting backgrounds from the input images, we can enhance the model’s focus on the speaker’s lip movements.
- **Visual-Geometry Information:** Instead of solely relying on visual information, we included lip landmarks to keep track of its motion trajectory.
- **Architectural Modifications:** Replacing the existing Bi-GRU [5] with a bidirectional LSTM [6] or transformer-based architectures [7] may help capture more complex temporal relationships in the data.
- **Attention Mechanisms:** Implementing attention mechanisms [7] can allow the model to prioritize and process the most relevant parts of the lip movements, improving accuracy.

The experimental results were commendable. Our best model of using 3D-conv image encoder with lip coordinate extractor with Bi-GRU and transformer performed the best with a CER of 2.78% and WER of 7.39%. Detailed ablation studies were performed to evaluate the importance of each component. Upon testing, we found that attention mechanism and lip landmarks are the key factors that makes LipCoordFormer standout.

2. Related Work

Our team has reviewed prior works in automated lip-reading, focusing on deep learning methods, convolutional sequence-to-sequence models and temporal modeling methods that are relevant to our research.

2.1. Automated Lip-Reading

Automated lip-reading must address two key challenges: extracting visual features from lip movements and mapping these features to spoken language. Early works employed model-based approaches, such as Tim et al. [8], who used normalized concatenated Active Appearance Models (AAM) features from consecutive frames to extract spatiotemporal features, and Luetttin et al. [9], who utilized the Active Shape Model (ASM) to generate features from key point coordinates. Alternative approaches used image-based features like 2D discrete cosine transform feature (DCT) [10] and articulatory features (AFs) [11], though these are more susceptible to environmental variations. For language mapping, traditional non-deep learning methods such as Hidden Markov Models (HMMs) [12] and Dynamic Bayesian Networks (DBNs) [13] were commonly used.

The emergence of deep learning combined these two challenges into one unified process through convolutional neural networks (CNNs). LipNet pioneered this approach by employing 3DCNNs for spatiotemporal feature extraction, making a significant leap in lip reading performance. Building of CNNs, subsequent methods like RNN-CTC [4], RNN-seq2seq [14], and Transformers [7] have further advanced automated lip-reading.

2.2. Convolutional sequence-to-Sequence Models

Sequence-to-sequence (seq2seq) models process variable-length input and output sequences. While Sutskever et al. [14] initially proposed seq2seq models with stacked RNNs for neural machine translation, convolutional seq2seq models (conv-seq2seq) using only convolutional layers in the encoder-decoder architecture have proven more effective for lip-reading due to their superior spatiotemporal feature capturing capabilities. For instance, LipFormer [15] employs 3DCNN with channel attention to extract visual features, which are then processed by a transformer encoder. Similarly, the Watch, Attend and Spell (WAS) network [2] combines 5-layer 2DCNNs with recurrent modules and attention mechanisms, using an LSTM transducer for decoding. Conv-seq2seq models have significantly advanced lip-reading performance and found widespread application in other domains [16]. Our work builds upon this foundation by integrating convolutional networks with transformer architectures while utilizing RNNs to enhance temporal feature extraction.

2.3. Temporal Modeling Methods

Temporal modeling remains critical for accurate lip-reading, as spoken language involves sequential dependencies. Zhang et al. [16] proposed using Spatio-Temporal Fusion Module (STFM) to condense high-dimensional CNN outputs into low-dimensional temporal features, and then using Temporal Focal blocks (TF-blocks) to capture short-range frame dependencies. Xu et al.'s LCA Net [17] combines an image encoder with a highway network and bi-GRU to capture both short-term and long-term temporal dynamics, using a cascaded attention-CTC decoder for text generation. These methods improve performance by better capturing the long-term temporal correlations inherent in lip movements.

Our work synthesizes these approaches by proposing an end-to-end model that combines CNNs, Transformers, and RNNs to capture the spatiotemporal features of lip movements.

3. Data

3.1. Dataset Choice and Information

We used the GRID dataset [18], which consists of video recordings of 34 subjects, each narrating 1,000 structured sentences. The dataset contains MP4-format videos, accompanied by corresponding textual transcripts, making it suitable for supervised learning. Each video contains one sentence in a fixed 6-word structure as "verb + color preposition + letter + number + adverb". While the dataset initially included 34,000 videos, the absence of speaker 21 and some corrupt or empty files reduces the usable videos to 32,746. Each video is 3 seconds long at 25 fps, with a resolution of 720×576 pixels. Due to memory limitations, we only used 8000 videos. Therefore, the total dataset now spans 6.67 hours with 8000 videos. We also utilized an 8:1:1 Train-Valid-Test split.

3.2. Dataset Justification

Our team has recognized that there are other datasets such as LRS2 (16GB) [2] and LRW (250GB) [2] that could also be used for this project. In fact, our team initially decided to use the LRS2 dataset and even received permission from the University of Oxford. However, after careful consideration of our computational resources and project goals, we have selected the GRID dataset for this project. While the LRS2 dataset provides diverse real-world speech samples, we have selected GRID due to its structured sentence format, ensuring clarity in lip-reading tasks. Although GRID's constrained vocabulary limits open-domain generalization, its controlled environment allows for precise speaker-dependent modeling. Furthermore, the sentence-level split method ensures robust evaluation across unseen speakers and varying sentence structures.

Additionally, our team has limited GPU resources, making GRID a more suitable choice for computational efficiency. With a dataset size of 16GB compared to LRS2's 50GB, GRID requires less storage and memory, easing the training process. The dataset's structured nature ensures more efficient model convergence, as it avoids unpredictable variability present in real-world video samples. Moreover, shorter video clips (3 seconds each) at 25 fps allow for easier processing and optimization. The consistent speaker setup and lighting conditions further reduce computational overhead compared to LRS2's complex scene variability. Finally, the GRID dataset has been widely used in most sentence prediction tasks for lip reading [1, 19, 20].

3.3. Additional Preprocessing

Facial landmark recognition is performed using Dlib [21], a library used for facial analysis. Dlib's pre-trained models accurately identify key landmarks around the mouth, allowing precise cropping of the lip region. All video clips will thus be cropped to include the lip only and resized to 120×120 pixels.

During initial training, we found that loading a MPEG file and converting each frame into tensor is time-consuming (nearly 20s for each data sample). Thus, we pre-compute each sample into tensor value before the training starts. Due to limited storage capacity, we quantize each video from FP16 to FP4, which significantly reduced the storage size of the GRID dataset.



Figure 1. Illustration of frame preprocessing. Upper: original frame; Middle: lip region highlighted; Bottom: cropped frames)

4. Methods

In this section, we introduce the deep learning architecture of the proposed LipCoordFormer. Our methodology combines spatiotemporal feature learning, transformer-based sequence modeling, and cross-modal alignment to address the limitations of prior lip-reading networks. Our model, shown in Figure 2, consists of 3 components, 1) Spatiotemporal Image Encoder, 2) Lip Coordinate Extractor, and 3) Coordinate-conditioned Transformer Decoder. The following figure is the overall architecture:

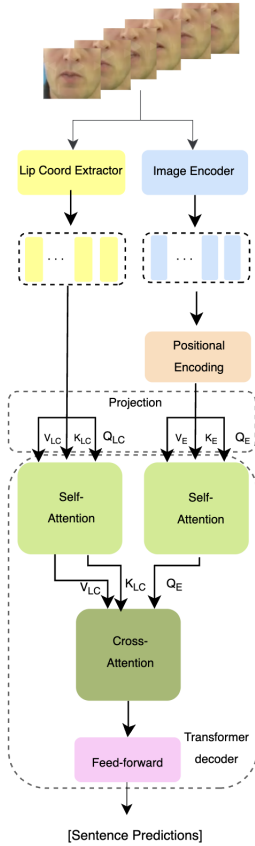


Figure 2. LipCoordFormer Overall Architecture

We start with a preprocessed lip video $X_{1:T} = [x_1, x_2, \dots, x_T]$ where $x_t \in \mathbb{R}^{3 \times H \times W}$ represents each image frame, and the ground truth sentence in natural language $S_{1:L} = [c_1, c_2, \dots, c_L]$ where $c_i \in D$ represent a single character from the vocabulary set

D , that is the 26 alphabets plus one blank space.

Preprocessed input frames are first passed through an image encoder to encode visual features into a sequence of continuous latent representations that preserves spatiotemporal features, denoted as $Z_{1:T} \in \mathbb{R}^{T \times C}$, where C equals to the hidden dimension of latent space, which is 512 in our case. On the left track, the input video frames are simultaneously passed through a lip landmarks extractor to extract the movements of lip regions represented in 20 X-Y coordinates, denoted as $LC_{1:T} \in \mathbb{R}^{T \times 20 \times 2}$. These geometric coordinates are then mapped to a latent embeddings $E_{1:T} \in \mathbb{R}^{T \times 2C}$.

Finally, these visual features and geometric coordinates are then passed into a transformer decoder, where the landmarks attend weights to focus on visual lip motions to achieve visual-landmark-temporal fusion. The output texts from transformer, $Y_{1:L'} = [y_1, y_2, \dots, y_{L'}]$, are then compared with ground truth texts using CTC loss.

4.1. Spatiotemporal Image Encoder

The encoder generates spatiotemporal latent representation for the input frame set $X_{1:T}$. This aims to produce a sequence of features that not only extract information from frame-level but align with temporal features bidirectionally. The encoder consists of 2 components: a CNN-based feature extractor and a bi-directional RNN module. For each of them, we explored 2 state-of-the-art architectures.

4.1.1 CNN-based Feature Extractor

1. **3D-ConvNet.** The first approach (Figure 3) is referenced from the state-of-the-art LipNet approach. The module composes of three layers of 3D convolutional blocks, each uses 128 filters with square kernel of 5x5, then followed by batch normalization, activation and pooling layer. This architecture is adopted due to its efficiency in achieving promising result for LipNet while having a light architecture [1].

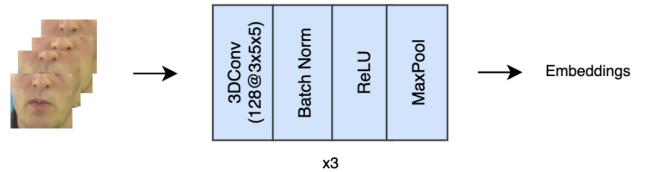


Figure 3. Original LipNet ConvNet Architecture

2. **3D-ResNet34.** We additionally explore 3D-ResNet34 [22] (with pooling and feed-forward layers removed) to evaluate whether dynamic and deeper architectures with residual connections enhance spatiotemporal feature extraction (Figure 4). It contains 34 layers of 3D convolution blocks with residual connections, starting with square kernels of 7x7 and perform consistent temporal down sampling. Compared with 3D-Conv, its larger kernel size expands spatial receptive field to potentially capture broader lip movements across frames. Also, ResNet34 uses structured temporal downsampling instead of the previous max pooling layers which may help to preserve information.

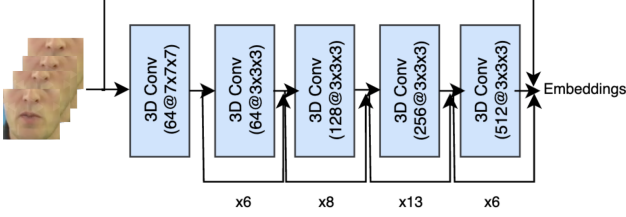


Figure 4. 3D-ResNet34 Architecture

By comparing these architectures, we test whether hierarchical residual learning improves temporal alignment and robustness to visual noise—key limitations in LipNet’s original design.

4.1.2 Temporal Modeling Unit

Given an input sequence of spatial features, $X' = [x'_1, x'_2, \dots, x'_T]$, we used a recurrent module to capture temporal features across frame to achieve visual-temporal fusion. Here we explored Bi-GRU [5] and Bi-LSTM [6].

4.1.3 Bi-GRU

The Bi-GRU computes forward $\vec{h}_t$ and backward $\overleftarrow{h}_t$ hidden states at each timestep by

$$\vec{h}_t = GRU(x'_t, \vec{h}_{t-1}) \quad (1)$$

$$\overleftarrow{h}_t = GRU(x'_t, \overleftarrow{h}_{t+1}) \quad (2)$$

where the GRU cell employs reset r_t and update z_t gates:

$$r_t = \sigma(W_r[x_t || h_{t-1}]) \quad (3)$$

$$z_t = \sigma(W_z[x_t || h_{t-1}]) \quad (4)$$

$$\tilde{h}_t = \tanh(W_h[x_t || (r_t \odot h_{t-1})]) \quad (5)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (6)$$

with W_r, W_z, W_h as learnable weights. The final state $h_t = [\vec{h}_t || \overleftarrow{h}_t]$ concatenates both directions. The embeddings is generated by stacking these operations.

4.1.4 Bi-LSTM

In contrast to Bi-GRU’s two-gate design, Bi-LSTM introduces a cell state c_t and three gates, that is, input i_t , forget f_t , and output o_t , to explicitly manage long-term dependencies. The cell state acts as a memory buffer, enabling precise control over information retention across extended sequences.

4.2. Lip Coordinate Extractor

Differ from state-of-the-art approaches [1, 2, 3, 23], we not solely depend on visual-temporal features, but also taken lip landmark into account. Visual representations of lip motion videos exercise large variance due to the different in lip color, shape, or the pouncing habit of a speaker [20]. Although GRID dataset is a high-quality dataset where the lip positions, movements, and facial expressions of speakers are strictly controlled, this does not

add robustness to our model as we land it to daily usage. To reduce variation, we utilized a Lip Extractor to generate a sequence of X-Y coordinates representing the lip movement for each video frame. Dlib’s 68-point facial landmark was used to identify the 20 points corresponding to the outer and inner lip contours. The lip landmarks are normalized to a range of [0,1] relative to the frame dimensions to ensure invariance to the resolution of the input video. For edge cases like failed landmark detection, we apply zero-paddings. As such, $LC_{1:T} \in \mathbb{R}^{T \times 20 \times 2}$ represents the motion trajectory of lip movement.

They were then passed through a recurrent layer to generate latent embeddings $E_{1:T} \in \mathbb{R}^{T \times 2C}$. Same as the above Image Encoder, both Bi-GRU and Bi-LSTM are tested. The whole pipeline is illustrated in Figure 5.

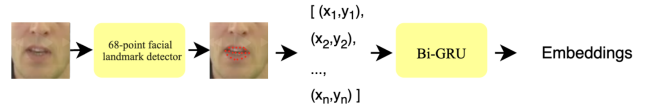


Figure 5. Lip Extractor Architecture

4.3. Coordinate-conditioned Transformer Decoder

The transformer accepts both embeddings from Image Encoder and Lip Coordinate Extractor. The decoder aligns lip movements with linguistic prediction by using both self-attention and cross-attention as shown in Figure 2. Positional embeddings are added to the visual features to encode temporal order. Self-attention operates on the encoded visual embeddings and lip landmark embeddings separately to capture temporal dependencies across frames. For a sequence of feature embeddings Z , the self-attention mechanism computes as follows,

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \quad (7)$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ are query, key, value matrices are derived from $\mathbf{Z}$

Cross-attention combines spatiotemporal feature $\mathbf{Z}$ with the lip embeddings $\mathbf{E}$ as follows,

$$\text{Cross-Attention}(\mathbf{Q}, \mathbf{K}_E, \mathbf{V}_E) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}_E^T}{\sqrt{d_k}}\right)\mathbf{V}_E \quad (8)$$

where $\mathbf{Q}$ is the query from $\mathbf{Z}$, and $\mathbf{K}_E, \mathbf{V}_E$ are the key and the value from $\mathbf{E}$. It takes visual features as queries ($\mathbf{Q}$) and geometric features as keys and values ($\mathbf{K}, \mathbf{V}$), where the latter features help to focus on discriminative regions by adjusting attention weights. In this pipeline, positional encoding mitigates frame-level ambiguity, self-attention enhances long-range temporal modeling, and cross-attention enables cross-modality learning.

4.4. Output

The output logits from the decoder represent a sequence of character probabilities, which are optimized by maximum likelihood training based on the CTC framework [4]. The objective

function to minimize is

$$\text{CTC Loss} = - \sum_{(\mathbf{x}, \mathbf{z}) \in \mathcal{S}} \ln(p(\mathbf{z}|\mathbf{x})) \quad (9)$$

Here, $p(\mathbf{z}|\mathbf{x})$ is the probability of the target sequence $\mathbf{z}$ given the input sequence $\mathbf{x}$, which is calculated by summing over all valid alignments between $\mathbf{z}_i$ and $\mathbf{x}_i$. This objective function also allows the model to learn from unaligned video-text pairs.

5. Experiments

In this section, we evaluate the performance of LipCoordFormer with ablation studies and compare it with other state-of-the-art lip-reading models also using GRID as training dataset.

5.1. Implementation Details

We implemented the model using approximately one fourth of the GRID dataset (8K videos), with each sample resized to 120×120 RGB frames at 25 Hz. We found that each sample has varying video length, so we dynamically padded each sample to synchronize input size. All videos are preprocessed as mentioned in Section 3. Training was conducted on an NVIDIA RTX 3080 GPU with 150 epochs and a batch size of 40 using Adam optimization. Learning rate is set to be $2e-5$ and dropout with rate 0.5 is activated. The total training time for each model was 12 hours.

Note that the initial settings in our progress report with LRS2 dataset and 50 epochs used to take 2 days to train, hence, to speed up and run all models, we shifted to a smaller GRID dataset.

5.2. Evaluation Metrics

We will be employing quantitative metrics using,

- **Word Error Rate (WER):** Measures the accuracy of predicted words against the ground truth.
- **Character Error Rate (CER):** Evaluates the accuracy at the character level.

5.3. Experiments

Methods	CER(%)	WER(%)
Hearing-Impaired Person (avg)	-	47.7
Lan et al. [24]	-	35.0
Wand et al. [19]	-	24.0
LipNet-LSTM [1]	15.2	26.3
LipNet-2DConv [1]	4.30	11.6
LipCoordFormer	2.78	7.49
LipNet-NoLM [1]	1.80	5.60
LipNet [1]	2.00	4.80

Table 1. Results of LipCoordFormer comparing to others

The performance of LipCoordNet in CER and WER comparing to other methods are summarized in Table 1. Upon testing the aforementioned variation in model architectures, the best model is

achieved with 3D ConvNet + Lip Landmark + Bi-GRU + Transformer. Our LipCoordNet achieved 2.78% CER and 7.49% of WER. Compared to LipNet, our model could out-perform 2 of its variations, including LipNet-LSTM and LipNet-2DConv. LipCoordNet is comparable and close to the performance of original LipNet, which is off by 0.78% CER and 2.69% WER, but only trained with 1/4 of its dataset. Compared with previous lip-reading deep learning models such as Lan et al. [24] and Wand et al. [19], we significantly reduced WER by over 4 times.

5.4. Ablation Studies

We conducted several ablation studies covering attention mechanisms, temporal modeling modules, and the Image Encoder. Unless specifically noted, all models are trained with the same experimental setup as mentioned in subsection 5.1.

5.4.1 Image Encoder

To evaluate the impact of the depth of visual encoder and residual learning, we compared using 3D-ConvNet with 3D-ResNet34 architecture. As mentioned, LipNet’s ConvNet are of much simpler architecture, without residual connections and maintain a constant kernel size. Yet, due to its rapid compression on feature maps, it failed to preserve fine-grained temporal details, while ResNet’s residual connection preserves spatial resolution. From Table 2, to our surprise, ResNet-based encoder recorded a performance degradation as compared to 3D-ConvNet. ResNet achieved a CER of 12.5% and WER of 44.3%, while 3D-ConvNet is about 6 times better than it. This may suggest that the increased model complexity led to overfitting on GRID’s limited vocabulary (6-word sentences), where simpler motions dominate. The 3D-ConvNet’s aggressive spatial pooling, while discarding fine-grained details, maybe more effective at compressing constrained lip patterns into robust features. We suspect this is a trade-off of using complex architectures, where deeper models require larger datasets to generalize effectively. However, more testing on larger dataset, e.g. the aforementioned LRS2 dataset, should be performed to prove this hypothesis.

Methods	CER(%)	WER(%)
LipCoordFormer-3DConv	2.78	7.49
LipCoordFormer-ResNet	12.5	44.3

Table 2. Ablation study on Image Encoder

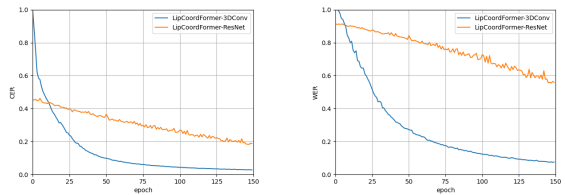


Figure 6. Validation curves of Ablation study on Image Encoder

5.4.2 Attention Mechanisms

One main difference between LipCoordFormer and LipNet is the use of attention mechanisms to align visual-geometric features. To evaluate the importance of cross-modality fusion, we compare each model variations with its transformer and positional encoding unit removed. Without attention, the models solely rely on convolutional and recurrent layers to process lip sequences. Results are shown in Table 3. This ablation led to a significant performance degradation. For LipCoordFormer with 3D ConvNet, CER increased by 2.82% and WER increased by 12.7% when transformer is removed. The same degradation for LipCoordFormer with ResNet34 variations. This gap underscores the critical role of attention in visual and geometric features fusions. By treating coordinates as keys and values, cross-attention guides the transformer decoder learns to dynamically weight regions of interest (e.g., lip corners, upper or lower lip contours), which helps to filter noises. Furthermore, self-attention enables the model to capture long-range temporal dependencies—such as co-articulation effects between syllables—that standard recurrent neural networks alone fail to model. These results validate our hypothesis that only having visual and geometric features is not sufficient for robust lip reading, which, the fusion between them is the key.

Methods	CER(%)	WER(%)
LipCoordFormer-3DConv	2.78	7.49
Without Transformer (3DConv)	5.60	20.2
LipCoordFormer-ResNet	12.5	44.3
Without Transformer (ResNet)	26.3	58.6

Table 3. Ablation study on Transformer (Both models use Bi-GRU and LipCoordFormer is with Transformer)

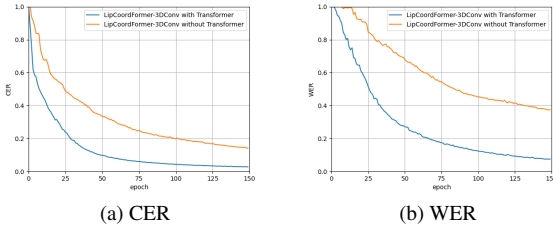


Figure 7. Validation curves of Ablation study on Transformer

5.4.3 Temporal Modeling Modules

The results of our comparison between Bi-GRU and Bi-LSTM reveal significant insights into the modeling of lip dynamics. We assessed the efficacy of Bi-GRU and Bi-LSTM, two of the most common recurrent models, in modeling lip dynamics. While Bi-GRUs use gated update mechanisms to process sequences bidirectionally, Bi-LSTMs introduce additional memory cells to track long-term dependencies. Keeping all other modules constant, we found that Bi-LSTM performed worse on our data. The findings

indicate that although Bi-LSTM is designed to capture more complex temporal patterns, it struggled with the nuances present in lip dynamics, leading to a higher Character Error Rate (CER) by approximately 50% (Table 4). This discrepancy suggests that the gating strategy of Bi-GRU, despite its simplicity, may be more effective for the sequential nature of lip movements. Theoretically, some papers suggest that the use of Bi-LSTM should lead to better performance; however, the main reason it does not perform well here is likely due to the limited amount of data, which leads to overfitting.

Methods	CER(%)	WER(%)
LipCoordFormer-3DConv-BiGRU	2.78	7.49
LipCoordFormer-3DConv-BiLSTM	57.3	98.7
LipCoordFormer-ResNet-BiGRU	12.5	44.3
LipCoordFormer-ResNet-BiLSTM	68.9	93.1

Table 4. Ablation study on Bi-GRU and Bi-LSTM (All with transformer)

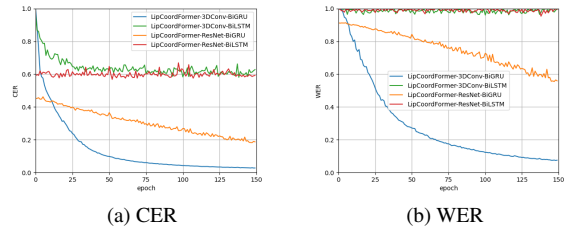


Figure 8. Validation curves of Ablation study on Bi-GRU and Bi-LSTM

6. Evaluation

6.1. Limitation

Despite these advancements, our research acknowledges certain limitations, including the model’s evaluation on a subset of the GRID dataset, restricting generalizability, and the potential for errors in the preprocessing step involving facial landmark detection. The reliance on the GRID dataset, with its structured sentence format and limited vocabulary, also limits the model’s ability to generalize to open-domain lip-reading tasks.

Additionally, the end goal is to use this model in a real-world context, such as integrating it into a wearable device. The model currently relies on accurate lip landmark detection, and its performance can be affected by errors in landmark localization. For example, in real-world scenarios, the model might struggle to interpret videos with challenging conditions like poor lighting or occlusions.

Furthermore, the current implementation does not explicitly model the co-articulation effects present in continuous speech. For example, the pronunciation of a phoneme is influenced by

its neighboring phonemes in continuous speech. This could limit the model’s ability to accurately decode more complex and natural speech patterns. Finally, the computational complexity of the transformer architecture, particularly the self-attention mechanism, can be a limiting factor for real-time applications, especially on resource-constrained devices.

6.2. Future Scope

In the future, we will focus on several key areas to further enhance the performance and robustness of LipCoordFormer. Given more memory resources, we plan to train the model with a significantly larger dataset to mitigate overfitting and improve generalization. We also intend to explore alternative, more powerful image encoder architectures, such as Vision Transformers (ViT) [25], to capture more complex visual features. Furthermore, we plan to investigate incorporating attention mechanisms within the coordinate extractor to dynamically weight the importance of different lip landmarks during the temporal modeling process. Finally, we aim to fine-tune LipCoordFormer on spoken language data to leverage the complementary information from audio signals, potentially leading to a more robust and accurate audio-visual speech recognition system.

7. Conclusion

In this study, we introduced LipCoordFormer as a novel approach to lip reading, addressing limitations in current state-of-the-art methods that rely on frame-level spatiotemporal convolutions and recurrent networks. Existing methods often struggle with visual noise from non-lip regions, inadequate modeling of long-range temporal dependencies, and visual-geometric alignments. LipCoordFormer distinguishes itself through a transformer-based architecture that integrates a spatiotemporal image encoder and lip movement landmarks. This integration enables the joint learning of visual-geometric patterns while eliminating frame-level alignment dependencies.

Through ablation studies, we explored different configurations of our model. We experimented with two image encoders: ResNet34 and ConvNet. Interestingly, the more complex ResNet34 did not perform as well as the simpler Conv3D, likely because the deeper network overfit the relatively small GRID dataset, failing to generalize effectively. Additionally, we investigated replacing the Bi-GRU with a Bi-LSTM in the lip coordinate extractor. However, the Bi-LSTM also underperformed compared to the Bi-GRU, potentially due to the increased complexity of the LSTM requiring more training data to optimize effectively.

Despite these variations, our core LipCoordFormer architecture demonstrated significant promise. Our experimental results demonstrate the effectiveness of this approach, with LipCoordFormer achieving a character error rate (CER) of 2.78% and a word error rate (WER) of 7.39% on the GRID corpus. This performance is comparable to existing state-of-the-art models (LipNet: CER: 6.4%, WER: 11.4%; LCArNet: CER: 1.3%, WER: 2.9%; STFM: WER: 1.3%). Importantly, LipCoordFormer achieves these results while introducing critical advancements in noise robustness and temporal dependency modeling, using only ¼ of the

training data, and reducing training time to 12 hours (compared to 7 days for STFM).

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